

=> Parametric Non-Parametric

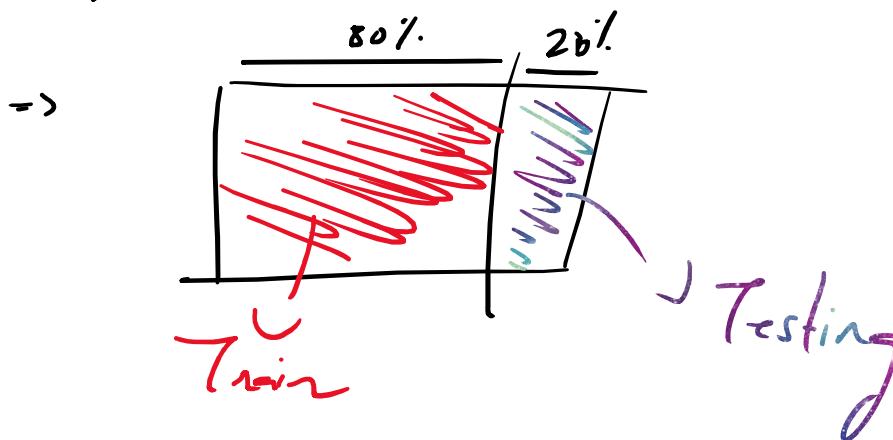
Pre-determine the
values from data.
(No optimization possible)

No pre-determination
possible.

$$\hat{y} = \beta_0 + (\beta_1 \cdot x_i)$$

=> Trying the accuracy:

- 1) Underfitting
- 2) Overfitting
- 3) Generalization

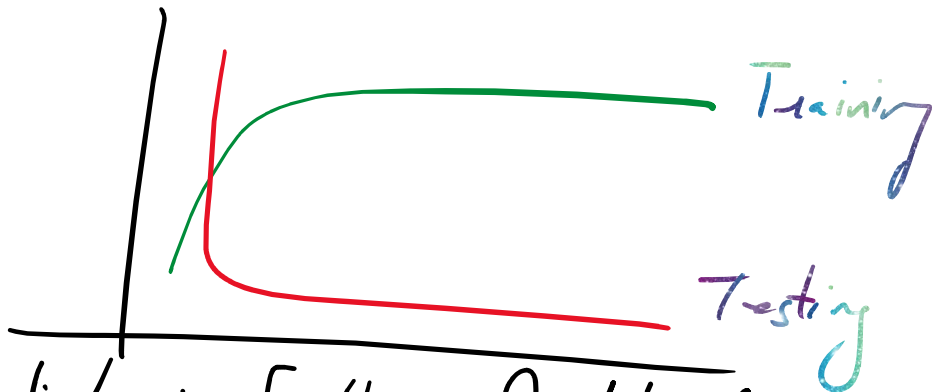


- i) Check the training samples (< 100k) | . . .

- => Underfitting: The model has capability but due to lack of data it does not perform well.

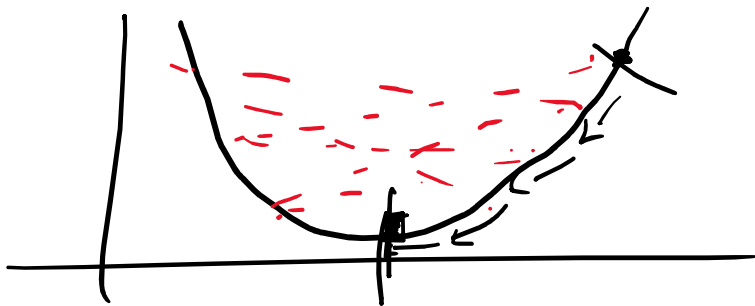


Overfitting: Abundance of data but something lacks in the model.



Generalization: Equilibrium of data & accuracies (τ_1/τ_2)

$\Rightarrow$ Optimization using Gradient Descent.



$$\hat{y} = \beta_0 + (\beta_1 \cdot x_i)$$

$$\begin{bmatrix} \frac{\partial}{\partial \beta_0} = A \\ \frac{\partial}{\partial \beta_1} = B \end{bmatrix} \quad \bigg| \quad \begin{bmatrix} \beta_0 - \frac{\partial}{\partial \beta_0} \\ \beta_1 - \frac{\partial}{\partial \beta_1} \end{bmatrix}$$

$\Rightarrow$ Subtract till when?

Epoch: Iterative procedure to run the optimizer.

as Subtract with what speed?

Learning Rate: (α): Determines the global minima is achieved without bouncing.
(0.0001 - 0.0005)

$\Rightarrow$ Implementation:

$$\text{loss} = \frac{1}{N} \sum (y_i - \hat{y}_i)^2$$

$$\frac{1}{N} \sum (y_i - \{\beta_0 + (\beta_1 \cdot x_i)\})^2$$

$$\Rightarrow \frac{\partial}{\partial \beta_1} = \frac{1}{N} \sum (y_i - \hat{y}_i)^2$$

$$\frac{\partial}{\partial \beta_1} = \frac{1}{N} \sum (y_i - \{\beta_0 + (\beta_1 \cdot x_i)\})^2$$

$$\frac{\partial}{\partial \beta_1} = \frac{1}{N} \sum 2 (y_i - \{\beta_0 + (\beta_1 \cdot x_i)\}) \times \frac{\partial}{\partial \beta_1} (y_i - \{\beta_0 + (\beta_1 \cdot x_i)\})$$

$$\frac{\partial}{\partial \beta_1} = \frac{2}{N} \sum (y_i - \{\beta_0 + \beta_1 \cdot x_i\}) \times (-x_i)$$

$$\frac{\partial}{\partial \beta_1} = -\frac{2}{N} \sum (x_i) (y_i - \{\beta_0 + [\beta_1 \cdot x_i]\})$$

$$\frac{\partial}{\partial \beta_0} = -\frac{2}{N} \sum (y_i - \{\beta_0 + (\beta_1 \cdot x_i)\})$$

$$\Rightarrow \beta_1 = \beta_1 - (\alpha \times \frac{\partial}{\partial \beta_1} \cdot \text{Cost Fn})$$

$$\beta_0 = \beta_0 - (\alpha \times \frac{\partial}{\partial \beta_0} \cdot \text{Cost Fn})$$

$\Rightarrow$ Epoch : {

$$\frac{\partial}{\partial \beta_1} \cdot \text{Loss Fn}$$

$$\frac{\partial}{\partial \beta_0} \cdot \text{Loss Fn}$$

$$\beta_1 = \beta_1 - (\alpha \cdot \frac{\partial}{\partial \beta_1} \text{Loss})$$

3

$$\beta_1 = \beta_1 - (\alpha \cdot \frac{\partial}{\partial \beta_1} \text{Loss})$$

$$\beta_0 = \beta_0 - (\alpha \cdot \frac{\partial}{\partial \beta_0} \text{Loss})$$

$$\hat{y} = \beta_0 + (\beta_1 \cdot x_i)$$

$$\hat{y} = \beta_0 + (\beta_1 x_i) + (\beta_2 x_j)$$

$$\hat{y} = \beta_0 + (\beta_1 x_i) + (\beta_2 x_j) + (\beta_3 x_k)$$

$\Rightarrow$ Least Squares Estimate : (Multivariate Regression)

$$\hat{b} = \begin{pmatrix} \hat{a} \\ \hat{b}_1 \\ \hat{b}_2 \\ \vdots \\ \hat{b}_n \end{pmatrix} = \left[(X^T \cdot X)^{-1} \cdot X^T \cdot y \right]$$

$\Rightarrow$ Bias : an arbitrary weight

Product 1 Sales Prod 2 sales Weekly Sales

1	4	1
2	5	6
3	8	8
4	2	12

$$X = \begin{pmatrix} 1 & 1 & 4 \\ 1 & 2 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 2 \end{pmatrix}$$

Bias

$$X^T X = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 5 & 8 & 2 \end{pmatrix} \times \begin{pmatrix} 1 & 1 & 4 \\ 1 & 2 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 2 \end{pmatrix}$$

$\boxed{3 \times 4}$ 4×3

$$= \begin{pmatrix} 4 & 10 & 19 \\ 6 & 30 & 46 \\ 19 & 46 & 109 \end{pmatrix}_{3 \times 3}$$

$$(X^T X)^{-1} = \begin{pmatrix} 3.15 & -0.59 & -0.30 \\ -0.59 & 0.20 & 0.016 \\ -0.03 & 0.016 & 0.054 \end{pmatrix}$$

$$\beta = (X^T X)^{-1} X^T y = \begin{pmatrix} 3.15 & -0.59 & -0.30 \end{pmatrix} \times \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 5 & 8 & 2 \end{pmatrix}$$

$$(X^T \cdot X)^{-1} \cdot X^T = \begin{pmatrix} 3.15 & -0.59 & -0.30 \\ -0.59 & 0.20 & 0.016 \\ -0.03 & 0.016 & 0.054 \end{pmatrix} \times \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 5 & 8 & 2 \end{pmatrix}$$

3×3 3×4

$$= \begin{pmatrix} 1.36 & 0.47 & -1.02 & 0.19 \\ -0.31 & -0.058 & 0.153 & 0.262 \\ -0.065 & 0.003 & 0.185 & -0.12 \end{pmatrix}_{3 \times 4}$$

$$((X^T \cdot X)^{-1} \cdot X^T) y = \begin{pmatrix} 1.36 & 0.47 & -1.02 & 0.19 \\ -0.31 & -0.058 & 0.153 & 0.262 \\ -0.065 & 0.003 & 0.185 & -0.12 \end{pmatrix} \times \begin{pmatrix} 1 \\ 6 \\ 8 \\ 12 \end{pmatrix}$$

(3×4) (4×1)

$$= \begin{pmatrix} -1.7 \\ 3.486 \\ 0.005 \end{pmatrix} \begin{matrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{matrix}$$

$$\hat{y} = -1.7 + 3.486 \times x_i + 0.005 \times x_j$$

$$P_1 = 5 \quad \& \quad P_2 = 3$$

$$\hat{y} = -1.7 + 3.486 \times 5 + 0.005 \times 3$$

$$= \boxed{15.74} \checkmark$$

$$\begin{pmatrix} \text{Bias} & 5 & 3 \end{pmatrix}_{1 \times 3} \times \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}_{3 \times 1}$$

Qx3

3 x 12